

Ethical Experience Design for the Value of Privacy based on Psychological Needs

Veronica Hoth
veronica.hoth@tu-berlin.de
Technische Universität Berlin
Berlin, Germany

Moritz Langner
moritz.langner@kit.edu
Karlsruhe Institute of Technology (KIT)
Karlsruhe, Germany

Anne Elisabeth Krüger
Anne-Elisabeth.Krueger@iao.fraunhofer.de
Fraunhofer Institute for Industrial Engineering IAO
Stuttgart, Germany

Stefan Brandenburg
sbranden@uni-mainz.de
Johannes Gutenberg-Universität Mainz,
Anwendungsorientierte Kognitionspsychologie
Mainz, Germany

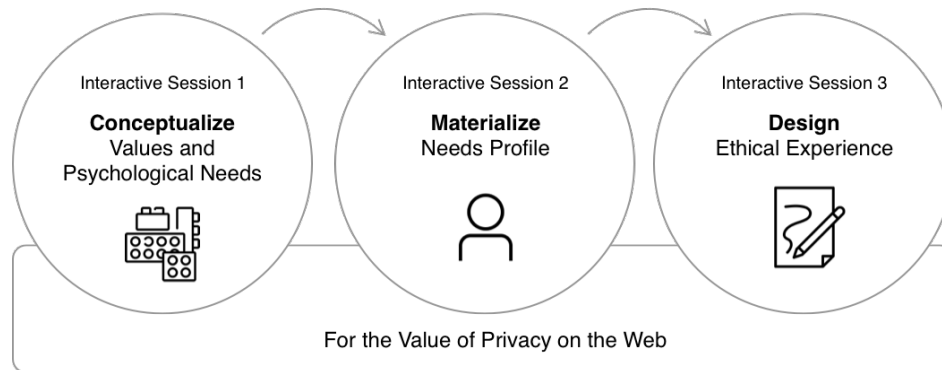


Figure 1: Course Content: After an introduction, the first interactive session will cover the concept of values and psychological needs in the context of Experience Design. The second interactive session focuses on how to materialize needs and values using the method Needs Profiles. The third interactive session covers an exemplary design process for creating an ethically sound and interactive experience. All sessions focus on the value of privacy on the web as an overall topic of this course.

ABSTRACT

Ethical design can avoid collateral damage by anticipating or counteracting possible negative consequences of technology on society, users, environment, among others. In this course, we aim to provide knowledge about psychological needs and values and the ethical implications of experience-centered interaction design. Therefore, the course consists of three interactive sessions. Each session includes briefings on the scientific background and approach, followed by interactive exercises and discussions so that participants can directly apply and deepen their knowledge. The course concludes with a summary of the learnings and a discussion about the applied methods. The results will be documented and subsequently made available to the participants.

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CCS CONCEPTS

• **Human-centered computing** → **User centered design.**

KEYWORDS

course, fundamentals of HCI, values, psychological needs, ethics, experience design

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1 BENEFITS

The human-centered design process (DIN EN ISO 9241-210) is an approach to designing products and services in an iterative and human-centered manner [7]. However, values, psychological needs, and ethical considerations are barely integrated across the various phases of the human-centered design process. Nevertheless, these

aspects are crucial for the user's experience with a product or service. Therefore, this CHI course teaches how to make the rather abstract values and psychological needs tangible and operationalize those concepts to investigate the ethical consequences of designing for experience. Participants will follow a guided process to design an exemplary cookie banner for the value of privacy and its related psychological needs.

At the beginning of the course, participants will be provided with the scientific background on values and psychological needs. Then, they will learn about the Needs Profiles method [12], which supports making tacit knowledge systematically tangible. The participants will apply the method of Needs Profiles, including Needs Personas [8], to make the abstract values and needs tangible, and to derive exemplary requirements and fuel the subsequent creative process of designing a cookie banner. Participants will use Consideration Cards [15] to identify and counteract possible negative consequences of experience design on users. Finally, participants will design a draft of a cookie banner considering the value of privacy and its related psychological needs as well as ethical consequences. Thus, this course will support researchers and practitioners in adopting an ethically considered, value- and needs-based human-centered experience design approach for their (research) projects, products, and services. Moreover, it is a platform for researchers and practitioners to share their experiences and knowledge in those areas.

2 INTENDED AUDIENCE(S)

The course is intended for researchers, students, and practitioners who:

- are interested in discussing the concepts and relevance of values and psychological needs in the context of experience design
- want to learn new methods to make the rather abstract concepts of values and psychological needs tangible and concrete, integrating tacit knowledge
- are confronted with ethical design challenges in their professional work
- are interested in co-creating new pathways to make ethical design decisions

3 PREREQUISITES

Knowledge about the human-centered design process (DIN EN ISO 9241-210) and a basic understanding about usability and user experience design are beneficial for course participants. Basic scientific knowledge about values and psychological needs will be introduced during the course. Therefore, no knowledge in this area is required. Furthermore, participants will engage with several new methods in interactive group exercises and should be open to share their opinion and experiences with the group.

4 CONTENT

The course takes place over four teaching units á 75 minutes. It starts with a presentation outlining the course's timeline, content, and potential learnings as well as an introduction to the concept of values and psychological needs [2, 4, 5] in the context of experience design based on literature. The focus is put on the value of privacy

and on how to integrate ethical evaluations in the human-centred design process. Then the three interactive sessions start (figure 1), which build on earlier successful workshops [10] and tutorials [6] that were combined and further developed. The course ends with a summary and a discussion about the applied methods, insights and results of the exercises.

4.1 Interactive Session 1 "Conceptualize" - Concepts of Values and Psychological Needs (75 min)

To prepare the participants for the topic of values and psychological needs in the context of experience design and to ask them about their previous experiences, the modified Warm UP "Object Presentation" [9] is used (Interactive Exercise 1). The participants are asked to individually build their understanding of the terms "psychological need" and "value" with Lego® bricks. The results are then shared and discussed with the other course participants.

Afterwards, a short presentation introduces the concept of values in the context of experience design based on literature. Then, the focus is put on the value of privacy. Now, a short presentation on psychological needs in the context of experience design is given [2, 4, 5]. Thereby, the participants listen actively. This means that they simultaneously listen, reflect and write down on sticky notes which psychological needs are relevant to them in the context of the value of privacy. The results are documented, consolidated, and discussed in the second interactive exercise.

4.2 Interactive Session 2 "Materialize" - Needs Profiles for the Value of Privacy (75 min)

In the second interactive session, the rather abstract psychological needs and values are made tangible and concrete. For this purpose, Needs Profiles from Krüger et al. [12] are applied to the privacy value. The Needs Profiles method can be used to systematically access implicit knowledge about psychological needs and their context [13], including values. The first step of the Needs Profiles method is to build the previously defined psychological needs metaphorically using Lego® bricks in the sense of the Lego® SeriousPlay® method. The insights gained during this third interactive exercise are then transferred to the Needs Empathy Map [8], and Needs Personas for the privacy value are developed as the second step for the Needs Profiles [8, 11, 12]. In this way, participants elaborate on how individuals with the selected psychological needs would act and behave in the context of the value of privacy and what their motivations, thoughts, and feelings are. Finally, the findings of the second interactive session are summarized in a plenary session. Here, it is discussed which mindset or attitude the participants should have to be able to design interactions in a value- and needs-oriented manner.

4.3 Interactive Session 3 - "Design" - Ethical Experiences for the Value of Privacy (150 min)

In the third session participants engage in a creative process by following three interactive exercises to apply the knowledge acquired in the previous session to sketch a new ethical design solution for

privacy interactions on the web. After a brief description of the topic, the participants will (a) define a requirement statement based on the Needs Profile, (b) conduct an ethical evaluation of the defined requirement and (c) visualize a new ethical design solution through idea sketching.

The fourth interactive exercise (a) of the design session builds upon the Needs Profile for the privacy value that was defined in session two. The participants specify the user requirements of the interaction with privacy dialogue boxes (i.e., cookie banners) by defining one specific problem in the form of a problem statement [1, 14]. This helps them to narrow down the more general knowledge about the user needs identified in the previous session to one specific user, insight and situation.

In the next interactive exercise (b), participants analyze the ethical implications the problem statement may have for the users by conducting an ethical evaluation exercise. This exercise was inspired by the “consideration cards” of Zhou [15] and has been modified by adding the values of Friedman et al. [3]. In the last interactive exercise (c), participants sketch a new interaction design solution for privacy on the web, taking into account the user requirements and the ethical implications. They will later present their design ideas in the plenary.

5 PRACTICAL WORK

The practical work of this course covers interactive sessions, group discussions, individual work and reflection phases. Participants work with templates, physical card sets, and different sets of Lego® bricks provided by the course instructors.

6 INSTRUCTOR BACKGROUND

Veronica Hoth is a PhD student at the Technical University of Berlin. She has over ten years of practical experience as a designer and specializes in Experience and User Centered Design. Her main research interest lies in implementing ethical design evaluations into the Human Centered Design process, using value sensitive design as a reference to increase privacy and diversity in HCI.

Anne Elisabeth Krüger is an (user) experience researcher at the Fraunhofer Institute for Industrial Engineering. She is writing her doctorate on creativity and empathy in the context of experience design. She is a trained Design Thinker (d.school) and Lego® SeriousPlay® facilitator as well as a value-based systemic coach and consultant (CAS/ISO/ICI). Furthermore, she focuses on enabling positive experiences in creative workshops in analog, digital, and hybrid settings.

Moritz Langner is a PhD student at Karlsruhe Institute of Technology (KIT) in Germany. His research interests focus on the application of eye-tracking technology in user-adaptive systems and the implications of eye-tracking technology on the privacy of users.

Stefan Brandenburg is currently a Professor of Applied Cognitive Psychology at Johannes Gutenberg Universität Mainz. Before, he was a postdoctoral researcher at TU Berlin. Here, he founded the first Institutional Review Board in 2009, that he was the chair of for ten years. His research interests include HMI design and ethical aspects of new technology. He has published on general ethical aspects of new technology and methods addressing these aspects in journals with peer review, magazines and at conferences.

7 RESOURCES

In the interactive sessions we will present the theoretical basis for the exercises. For the interactive exercises, we will provide templates for the course participants to use. We can provide a list of references in addition to the course material for those who want to deepen their knowledge on ethical design, value sensitive design and psychological (user) needs after the course.

8 ACCESSIBILITY

We improve the accessibility of our workshop by providing course material at different heights (e.g., on pinboards) and by physical assistance via student volunteers that we will bring to the conference. He or she can explain the proceedings of the course or course material with visual impairments. In case of an online course, we want to use automatic zoom transcriptions for people with auditory impairments.

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